

FULL STACK APPLICATION DEVELOPMENT

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DEPARTMENT OF CSE, CS&IT, AI&DS
COURSE CODE: 24SDCS02 / 24SDCS02E / 24SDCS02P / 24SDCS02L
FULL STACK APPLICATION DEVELOPMENT

Date of the Session: __/__/____

Time of The Session: ____to____

#SKILL - 5 → Spring Autowiring Demo using @Autowired

Prerequisites:

- General Idea on Spring Framework
- Modules of Spring Framework
- Understanding of Autowiring concepts

A college application maintains **student** information along with their **certification details**. The **Student** object contains a **Certification** object, and instead of creating this Certification object manually, Spring should automatically provide it using the **@Autowired** annotation. When the required classes are marked with **@Component**, Spring creates their objects and manages them inside the **IoC container**. Using **@Autowired**, the Certification object is supplied to the **Student** object by Spring without using the **new keyword**, allowing both objects to be linked through **annotation-based configuration**.

Tasks:

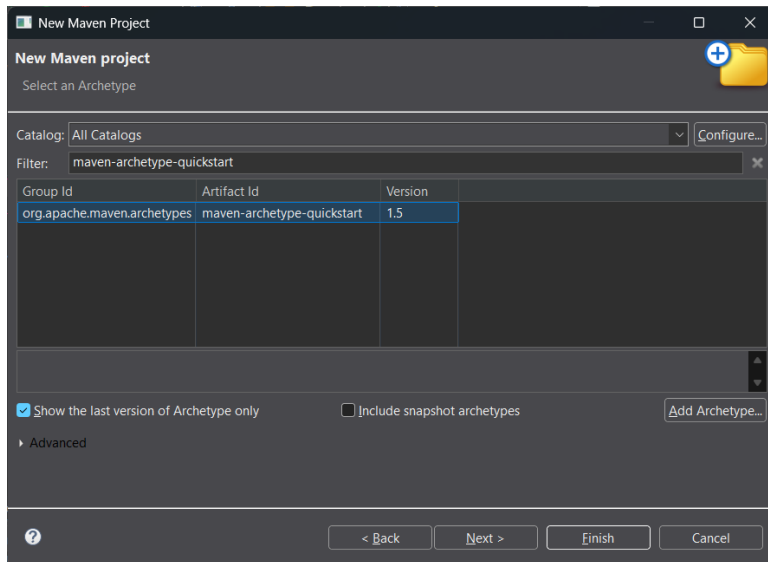
1. Create a **Certification** class with fields such as id, name, and dateOfCompletion.
2. Create a **Student** class with fields such as id, name, gender, and a Certification object.
3. Annotate both classes with **@Component** to enable Spring detection.
4. Use **@Autowired** on a field, constructor, or setter inside the Student class to inject the Certification object.
5. Configure component scanning using either:
 - a. An XML configuration file
 - b. A Java configuration class with **@Configuration** and **@ComponentScan**
6. Load the Spring IoC container using **ApplicationContext**.
7. Retrieve the Student bean and print all details, including the injected Certification object.
8. **Push the Spring project to GitHub.**

Aim:-

To demonstrate Spring Autowiring using the @Autowired annotation by injecting a Certification object into a Student object using annotation-based configuration.

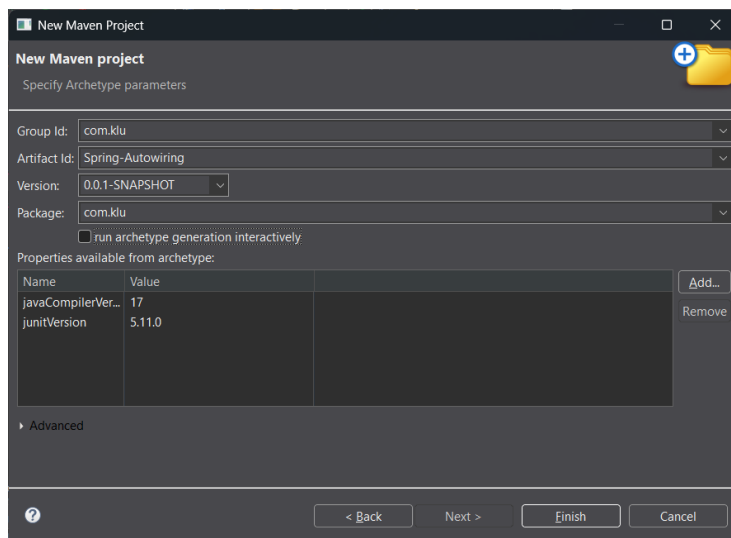
❖ Prerequisites:-

- Basic knowledge of Spring Framework
- Understanding of Spring modules
- Knowledge of Autowiring concepts



Enter details:

- **Group Id:** com.klu
- **Artifact Id:** Spring-Autowiring
- **Packaging:** com.klu
- **Finish**



❖ Project Structure:-

com.klu

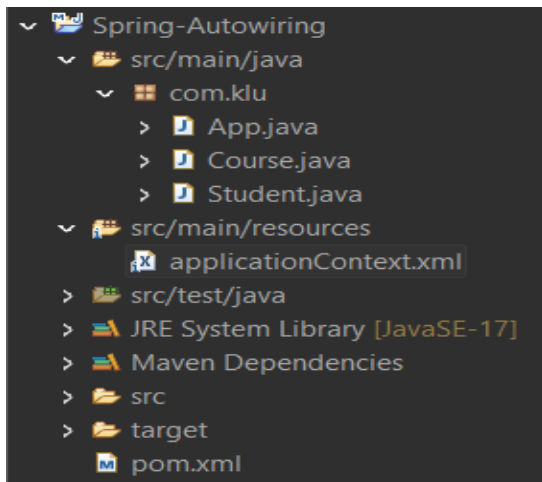
├─ App.java

├─ Student.java

└─ Certification.java

src/main/resources

---applicationContext.xml



Add Spring Dependency (pom.xml):-

```
<!-- Spring Context -->
<dependency>
  <groupId>org.springframework</groupId>
  <artifactId>spring-context</artifactId>
  <version>5.3.30</version>
</dependency>
```

Course.java:-

```
package com.klu;
```

```
import org.springframework.stereotype.Component;
```

```
@Component
```

```
public class Course {
```

```
    private int id = 301;
```

```
    private String course = "Spring Core";
```

```
    private String dateOfCompletion = "25-Jan-2026";
```

```
@Override
```

```
public String toString() {
```

```
    return "Course [id=" + id +
```

```
        ", course=" + course +
```

```
        ", dateOfCompletion=" + dateOfCompletion + "]);
```

```
}
```

```
}
```

Student.java:-

```
package com.klu;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Component;

@Component
public class Student {

    private int id = 10;
    private String name = "Divya";
    private String gender = "Female";

    @Autowired
    private Course course;

    public void display() {
        System.out.println("Student ID : " + id);
        System.out.println("Student Name : " + name);
        System.out.println("Gender : " + gender);
        System.out.println("Course : " + course);
    }
}
```

applicationContext.xml

 src/main/resources/applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns:context="http://www.springframework.org/schema/context"
    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
        https://www.springframework.org/schema/beans/spring-beans.xsd
        http://www.springframework.org/schema/context
        https://www.springframework.org/schema/context/spring-context.xsd">

    <context:component-scan base-package="com.klu"/>

</beans>
```

App.java:-

```
package com.klu;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class App {

    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

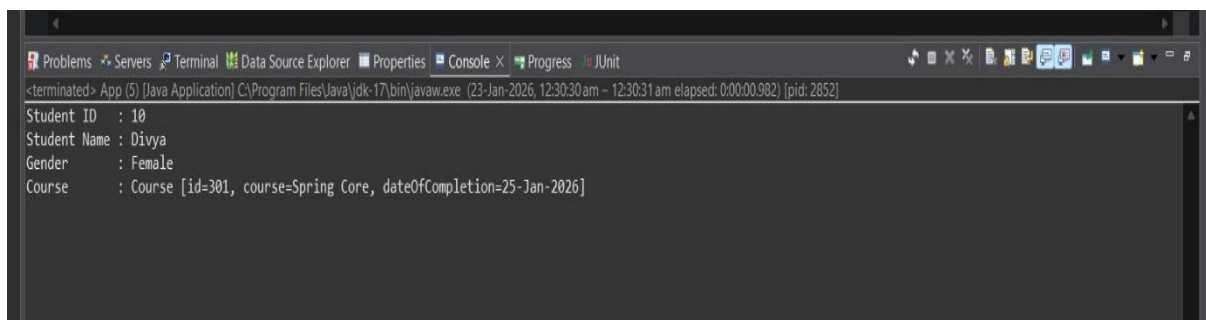
        Student student = context.getBean(Student.class);
        student.display();
    }
}
```

=====

Run the Application:

1. Right-click App.java
2. Run As → Java Application

OUTPUT on the console:



```
<terminated> App (5) [Java Application] C:\Program Files\Java\jdk-17\bin\javaw.exe (23-Jan-2026, 12:30:30 am - 12:30:31 am elapsed: 0:00:00.982) [pid: 2852]
Student ID : 10
Student Name : Divya
Gender : Female
Course : Course [id=301, course=Spring Core, dateOfCompletion=25-Jan-2026]
```

8. Push the Spring project to GitHub.

```
git init
git add .
git commit -m "Spring Autowiring Demo using @Autowired"
git branch -M main
git remote add origin https://github.com/your-username/Spring-Autowiring-Demo.git
git push -u origin main
```